ANEW Forklift Training Manual

Complete Course Materials

Powered Industrial Truck Training

Instructor: Mandy Richardson
Email: Mandy@anewcareer.org
Phone: 206-710-1019

ANEW - Building People, Building Communities

Table of Contents

1. Introduction and Safety Overview (Slides 1-10)
2. Employer and Employee Responsibilities (Slides 11-15)
3. Powered Industrial Truck Types and Classes (Slides 16-40)
4. Forklift Components and Differences (Slides 41-53)
5. Pre-Operational Checks and Inspections (Slides 54-60)
6. Load Handling and Stability (Slides 61-74)
7. Safe Operation and Rules of the Road (Slides 75-82)
8. Loading Docks and Parking (Slides 83-87)
9. Attachments and Equipment-Specific Training (Slide 88-89)
10. References

Section 1: Introduction and Safety Overview

Slide 1: Title Slide

ANEW Powered Industrial Truck Training

This comprehensive training program meets OSHA requirements under 29 CFR 1910.178 for powered industrial truck operator certification. Training includes formal instruction, practical hands-on training, and performance evaluation.

Slide 2: Forklift Safety - Why It Matters

Key Safety Message: "Of all the things that are important to you, where would you rank safety? If you have a family, you want them to be safe. You want to be safe while you're at home or when you're at work."

Critical Statistics:

- Forklifts weigh approximately 9,000 pounds; with a load, they can weigh up to 14,000 pounds (equivalent to six cars)
- OSHA estimates 68,000 forklift accidents occur annually in the US
- Over 100 fatalities occur each year
- 35,000 serious injuries and 62,000 non-serious injuries annually
- 11% of all forklifts are involved in accidents every year

Accident Breakdown by Cause:

- 25.3% - Tip over incidents
- 18.8% - Personnel struck by forklift
- 14.4% - Personnel struck by falling load
- 12.2% - Employee falls off elevated forks
- 7% - Forklift runs off loading dock or surface
- 6.1% - Improper maintenance

The Cost of Accidents:

- Lost time and productivity
- Increased insurance premiums
- Litigation costs (accidents commonly cost companies over $1 million)
- Personal injury and family impact
- Some companies cannot survive and close permanently

The Solution: Government studies show effective training programs improve forklift operator safety performance by as much as 70%. OSHA requires proper training for all lift truck operators, and only trained and authorized personnel can operate specific equipment[2][3][12].

Slide 3: Real-World Incident - Moped Collision

Case Study: Woman Rides Moped Into Forklift Prongs

This incident demonstrates critical safety failures:

- Forklift driver was distracted, looking at cell phone
- Victim on electric moped slammed into forklift forks
- Victim sustained head injuries from impact with floor

Lessons Learned:

- **Pedestrians always have right of way**
- No cell phone use while operating forklifts
- Importance of designated traffic lanes and floor markings
- Need for constant situational awareness
- Pedestrian safety training is essential

Slide 4: Fatal Accident Example - Three P's

Case Study: Heavy Iron Sheets Accident

What Happened:

- Workers attempted to move heavy iron sheets knowing they were too heavy
- One worker squatted on back of forklift as counterbalance
- Victim tried to help by pulling forklift down
- Victim slipped and fell under the forklift
- Metal sheets slid off forks and forklift crashed down on victim

Critical Failures:

- Exceeding rated capacity
- Unauthorized rider on forklift

- Improper load handling procedures
- No proper rigging or load assessment

Three P's to Safeguard:

1. **Personnel** - Operator and pedestrian safety
2. **Property** - Equipment and facility protection
3. **Product** - Load integrity and damage prevention

Slide 5-8: Accident Statistics

Fatal Accident Types:

- **42%** - Crushed by vehicle tipping over (leading cause of death)
- **25%** - Crushed between vehicle and surface
- **10%** - Struck or run over by forklift
- **8%** - Struck by falling material from forklift
- **15%** - Fall from elevated platform/forks

Annual Accident Totals:

- 85 fatal accidents per year
- 61,800 non-serious accidents per year
- 34,900 accidents resulting in serious injury

Non-Fatal Injury Distribution:

- Fractures: 22%
- Bruises and contusions: 20%
- Sprains and muscle tears: 19%
- Soreness and pain: 19%
- Cuts, lacerations, punctures: 5%
- Fractures with other injuries: 1%
- Amputations: 1%
- Sprains with other injuries: 2%
- All other: 11%

Understanding Non-Fatal Injuries:

While fatalities capture attention, non-fatal injuries result in significant costs: lost work time, increased insurance premiums, reduced productivity, and medical expenses. Common patterns include:

- **Fractures (22%):** Result from falls, tip-overs, or being struck by forklift/falling loads
- **Bruises/contusions (20%):** From being pinched, struck, or caught between forklift and fixed objects
- **Sprains/tears (19%):** Common when operators jump from forklifts or experience sudden stops
- **Soreness/pain (19%):** Cumulative trauma from vibration, poor posture, repetitive motions

Prevention: Proper training, adherence to procedures, seatbelt use, three-point contact when mounting/dismounting, and never jumping from equipment significantly reduce injury rates[20][23][31][33].

Slide 9-10: Hazard Identification

Physical Hazards:

- Blind spots around mast, overhead guard, and loads
- Narrow aisles and congested work areas
- Uneven floors, potholes, debris
- Wet, icy, or slippery surfaces
- Overhead obstructions (pipes, lights, doorways)
- Edge-of-dock and loading platform fall risks
- Trailer creep and early departure from docks

Operational Hazards:

- Tip-over risks from overloading, unbalanced loads, excessive speed
- Struck-by incidents involving pedestrians and equipment
- Falling loads due to improper stacking
- Crush points between forklift and fixed objects
- Visibility obstructions from loads
- Improper attachment use or exceeding capacity limits

Maintenance and Equipment Hazards:

- Faulty brakes, steering, or hydraulics
- Worn or damaged forks, chains, and tires
- Malfunctioning lights, horns, and backup alarms
- Fuel leaks and fire risks during refueling
- Battery acid exposure and hydrogen gas during charging
- Missing or illegible data plates and safety labels

Human Factors:

- Inadequate or expired training
- Operator fatigue, distraction, or impairment
- Failure to wear seatbelts
- Horseplay and unsafe behaviors
- Poor communication between operators and ground personnel

Environmental Hazards:

- Poor lighting in operating areas
- Excessive noise levels affecting communication
- Temperature extremes affecting equipment and operator performance
- Inadequate ventilation leading to exhaust fume buildup

Sources: OSHA hazard identification guidelines[20][21][23][29][31][33]

Section 2: Employer and Employee Responsibilities

Slide 11: Responsibilities of an Employer

Employers are legally required to provide:

- Comprehensive training programs meeting OSHA standards
- Safe working environment
- Right equipment for the job
- Sound, mechanically operable equipment
- Site-specific and equipment-specific training
- Performance evaluations and documentation
- Refresher training when needed

Slide 12: Responsibilities of an Employee

Employees must:

- Eliminate personal injury risk through safe practices
- Prevent product damage through proper handling
- Prevent property damage through careful operation
- Reduce maintenance costs by following procedures
- Provide a safe working environment for fellow employees
- Report unsafe conditions immediately
- Follow all training and operational procedures

Slide 13-14: Training Requirements

OSHA Mandate (29 CFR 1910.178(l)):

Training must be based on performance and tailored to fit the workspace and vehicle types. Key requirement: **No one under 18 can drive a forklift** - this is federal OSHA law.

Program Components:

1. Formal Instruction (Classroom/Online):

- Vehicle capacity, stability, and load handling principles
- Operating instructions and controls
- Vehicle inspection procedures
- Hazard recognition and avoidance
- OSHA regulatory requirements
- Manufacturer specifications and limitations
- Workplace-specific hazards and conditions

2. Practical Training (Hands-On):

- Supervised operation in actual work environment
- Load manipulation, stacking, unstacking
- Maneuvering in narrow aisles and tight spaces

- Traveling with and without loads
- Parking and securing equipment
- Refueling/recharging procedures
- Operating on ramps and inclines

3. Performance Evaluation:

- Direct observation by qualified trainer
- Assessment in workplace conditions
- Verification of safe operating practices
- Documentation of competency

Trainer Qualifications:

Trainers must have knowledge, training, and experience to instruct and evaluate operators. This can be demonstrated through recognized degrees, certificates, professional standing, or documented expertise in forklift operation and safety[32].

Training Documentation Requirements:

- Operator's name
- Date of training
- Date of evaluation
- Trainer's identity
- Type of training provided (initial, refresher, or evaluation)

Refresher Training Required:

- At least every 3 years (performance evaluation)
- After an accident or near-miss
- When unsafe operation observed
- When assigned to different truck type
- When workplace conditions change

Sources: OSHA training requirements[12][14][15][16][27][32]

Slide 15: Operator Qualifications

Minimum Qualifications:

- Must be at least 18 years old (federal OSHA requirement)
- No motor vehicle driver's license required by OSHA (employer may require)
- Physically capable of operating equipment safely
- Able to read and understand operating instructions and safety procedures
- Free from conditions that could impair safe operation

Training Requirements:

- Complete formal classroom instruction covering truck-related and workplace-related topics
- Receive hands-on practical training in actual workplace conditions
- Successfully pass performance evaluation by qualified trainer
- Obtain employer certification documenting training completion

- Receive site-specific and equipment-specific training

Medical/Physical Considerations:

- Good vision (corrected vision acceptable)
- Adequate hearing (to hear warnings, alarms, and communication)
- Physical ability to operate controls and maintain alertness
- No conditions causing sudden loss of consciousness or control
- Not under influence of drugs, alcohol, or impairing medications

Prohibited Conditions for Operation:

- Under 18 years of age
- Not trained and certified for specific equipment type
- Under influence of alcohol or drugs
- Physically or mentally impaired
- Operating without authorization
- Has not completed refresher training when required

Sources: OSHA operator qualification standards[3][15][22][24][28][30]

Section 3: Powered Industrial Truck Types and Classes

Slide 16: Powered Industrial Truck Definition

Definition: A mobile, power-propelled truck used to carry, push, pull, lift, stack or tier materials. Powered by electric or combustion engines.

Also Known As: Forklifts, pallet trucks, rider trucks, fork trucks, or lift trucks.

Excluded: Vehicles used for earth moving and over-the-road hauling are not considered powered industrial trucks.

What Makes a P.I.T.:
Any kind of lifting device or material handling equipment that is powered by an engine (combustion or electric). They carry, push, pull, lift, stock, or tier materials.

These are NOT Powered Industrial Trucks:

- Skid steers (adding forks doesn't make it a forklift)
- Farming equipment
- Non-powered pallet jacks
- Bobcats with buckets

Safety Context:

"While forklifts make our jobs easier, they can also be very dangerous. More than 100 workers are killed and thousands severely injured in forklift accidents every year. The people who get hurt include both pedestrians and forklift operators themselves. In addition to the human cost, there's also the damage these accidents do to materials, facilities and the forklifts themselves."

Slide 17-20: Types and 7 Classes of PIT

Four Main Types:

1. **Counterbalanced** - Counterweight at rear offsets load at front
2. **Narrow Aisle** - Trilateral/bilateral turret trucks for aisles 1,500-1,800mm
3. **Pallet Truck** - Most basic form, moves pallets in warehouse/trailer
4. **Aerial Equipment** - Provides temporary access to inaccessible areas over 2 meters high

7 Classes of Powered Industrial Trucks:

- **Class 1:** Electric Motor rider truck
- **Class 2:** Electric Motor narrow aisle trucks
- **Class 3:** Electric Motor hand trucks or hand/rider trucks
- **Class 4:** Internal combustion engine trucks (solid/cushion tires)
- **Class 5:** Internal combustion engine truck (pneumatic tires)
- **Class 6:** Electric/internal combustion engine tractors
- **Class 7:** Rough terrain forklift trucks

OSHA Training Requirement: Operators must be trained and authorized for the specific class and type of forklift they will operate.

Slide 21-22: Class 1 - Electric Motor Rider

Description: Electric-powered counterbalance trucks with rider (seated or standing) positions. Most common indoor material handling vehicles.

Variants:

- Three-wheeled electric trucks (sit-down)
- Counterbalanced rider type, cushioned tires (high/low platform)
- Counterbalanced rider, pneumatic tire (high/low platform)
- Counterbalanced rider type, stand-up

Key Features:

- Electric motor with counterweight in rear
- Pneumatic (air-filled) or solid tires for puncture resistance
- Operator sits with seatbelt and rollover protection
- Telescoping mast picks up forks
- Typically 4-5 foot forks for warehouse use
- Battery-powered (zero emissions)
- Quiet operation
- Lower operating costs per hour

Typical Applications:

- Indoor warehouse operations
- Manufacturing facilities
- Loading docks
- Distribution centers
- Retail operations

Slide 23-26: Class 2 - Electric Motor Narrow Aisle Trucks

Description: Electric-powered narrow aisle equipment designed for high-density storage operations. Includes reach trucks, order pickers, and turret trucks.

Key Features:

- Narrower than Class 1 for tight aisle operation
- Solid, non-marking tires for warehouse floors
- Forks move up/down, in/out, and side-to-side
- Work in aisles 8-10 feet wide (VNA: 5-6 feet)
- Lift heights up to 50+ feet
- Man-up (operator rises with load) or man-down configurations

Safety Requirements:

- Do not have standard guardrails on all open sides
- Must be equipped with approved fall arrest system
- **Body belts are NOT acceptable fall protection**
- Full-body harness required for elevated operations
- Platform must not be raised when truck is in motion
- Operators must remain within platform confines

Applications:

- High-density warehouse storage
- Order picking and fulfillment
- E-commerce distribution centers
- Retail backroom inventory management

Sources: Narrow aisle specifications[55][58][61][64][67][70][73]

Slide 27-28: Class 3 - Hand and Hand Rider Trucks

Description: Electric-powered pallet jacks, stackers, and tow tractors for ground-level pallet movement and low-lift applications.

Includes:

- Electric pallet jacks
- Stackers
- Tow tractors
- Walk-behind ("walkie") models
- Rider models (operator stands on platform)

Key Features:

- Walk-behind or rider operation
- Limited lift height (typically 7-8 inches for pallet jacks)
- Handles with brakes and direct steering controls
- Perfect for unloading deliveries and moving loads to staging areas
- Commonly found in grocery stores and retail operations

Inspection Requirements:

- Safety devices
- Electrical systems
- Wheels
- Automatic stop function

Slide 29-31: Class 4 - Internal Combustion Engine Trucks (Cushion Tires)

Description: Internal combustion powered counterbalance trucks with solid cushion tires designed for indoor or smooth surface operations.

Key Features:

- Smooth cushion tires (solid rubber, no air)
- Very limited ground clearance (not suitable for outdoor/rough terrain)
- Cannot handle dirt, soft soil, rocks, or uneven surfaces
- Designed for indoor or controlled slab surfaces
- Usually LPG (propane), some diesel or gasoline
- Equipped with emission scrubbers for indoor use
- Quick refueling (2-5 minutes)
- Unlimited runtime

Limitations:

- Not suitable for outdoor rough terrain
- Limited ground clearance restricts surface conditions
- Cushion tires wear faster on abrasive surfaces
- Produce exhaust emissions (ventilation required)
- Noisy operation compared to electric

Applications:

- Indoor warehouses with smooth concrete floors
- Manufacturing facilities with finished floors
- Loading docks with level surfaces
- Distribution centers

Slide 32-34: Class 5 - Internal Combustion Engine Trucks (Pneumatic Tires)

Description: Internal combustion powered counterbalance trucks with pneumatic (air-filled) tires designed for indoor and outdoor operations, including moderately rough surfaces.

Key Features:

- Pneumatic tires with aggressive tread patterns
- Higher ground clearance (6-10 inches)
- Better shock absorption and traction
- Suitable for uneven terrain and moderate slopes
- Typically diesel, propane, or gasoline powered
- Available in capacities from 3,000 to 50,000+ lbs

- Air-filled or foam-filled tire options

Applications:

- Outdoor lumberyards
- Construction material yards
- Container handling in ports
- Outdoor storage yards
- Landscaping and nursery operations
- Railroad freight handling
- Mixed indoor/outdoor operations

Slide 35-36: Class 6 - Electric and Internal Combustion Engine Tractors

Description: Tow tractors and tuggers—specialized powered industrial trucks designed to pull carts, trailers, and material handling trains rather than lift loads.

Key Features:

- No lifting capability (no mast or forks)
- Designed exclusively for towing and pulling
- Can pull multiple connected carts or trailers
- Electric battery-powered or internal combustion
- Towing capacities from 2,000 to 100,000+ lbs

Applications:

- Manufacturing: pulling material trains between production stations
- Warehousing: moving pallet trains from receiving to storage
- Airports: luggage cart trains (baggage handling)
- Hospitals: linen, medical supply, food service, waste removal
- Retail: stock replenishment from backroom to sales floor

Slide 37-40: Class 7 - Rough Terrain Forklift Trucks

Description: Specifically engineered for outdoor operation on uneven, unpaved, and challenging surfaces where standard forklifts cannot function safely.

Key Features:

- Large pneumatic tires with deep, aggressive tread
- High ground clearance (12+ inches)
- Heavy-duty frame and suspension
- Four-wheel drive (4WD) available
- Diesel powered (most common)
- Telescoping boom extends forward and upward (telehandlers)

Types:

- **Rough Terrain Mast Forklifts:** Standard counterbalance for rough surfaces, vertical lift
- **Rough Terrain Reach Forklifts (Telehandlers):** Can reach 30-50+ feet high and 20+ feet forward
- **Specialized Variants:** Container handlers, log stackers, all-terrain forklifts

Applications:

- Construction sites
- Lumberyards and sawmills
- Agriculture and farming
- Mining and quarries
- Disaster relief and military operations

Safety Note: Load capacity decreases with boom extension and angle. Operators must understand load charts and capacity changes based on reach and height.

Section 4: Forklift Components and Differences

Slide 41-43: Components of a Forklift Truck

Engine Powered Forklift Components:

- Mast (telescoping vertical structure)
- Overhead guard (protective structure)
- Forks (load-bearing arms)
- Engine (power source)
- Front wheels (pivot point)
- Front axle
- Clutch or torque converter
- Rear wheels (steering)
- Transmission

Battery Powered Forklift Components:

- Mast
- Overhead guard
- Batteries (counterweight and power)
- Pump motor
- Hydraulic pump
- PS motor
- Forks
- Controller
- Front/rear wheels
- Front axle
- Drive motor
- Transmission

Critical Components Detail:

Mast Assembly:

- Interlocking vertical channels (2-3 stages)
- Lift cylinders (hydraulic extension)
- Tilt cylinders (forward/backward angle control)

- Lift chains (equal tension required)
- Rollers/bushings (guide mast stages)

Overhead Guard:

- Required by OSHA for most forklifts
- Protects operator from falling objects
- Does NOT protect from full-capacity falling loads
- Must be inspected for cracks, bends, missing bolts
- Houses lights, mirrors, accessories

Forks (Tines):

- Standard lengths: 42-48 inches (up to 96+ inches specialized)
- Load capacity stamped on fork
- Must be matched pairs (same length/capacity)
- Wear limit: when heel worn to 90% of original thickness, must replace
- Fork positioning should be 1/2 to 3/4 width of load

Sources: Forklift components[56][59][62][65][68][71]

Slide 44-45: Forklift vs. Automobile Differences

Critical Differences:

Feature	Automobile	Forklift
Steering	Front wheels	Rear wheels
Weight	3,000-5,000 lbs	9,000-20,000+ lbs
Center of Gravity	Low, centered	High, forward with load
Stability	Very stable	Naturally unstable
Speed	25-70+ mph	8-18 mph maximum
Visibility	360° good	Mast/load blocks view
Passengers	Multiple	Operator only
Braking	Front/rear	Typically rear-wheel only

Why These Differences Matter:

"Contrary to what many people think, driving a forklift is not like driving a car. Forklifts have different controls, they are heavier than cars, they steer with their rear wheels, and they are naturally unstable. Forklifts are designed to raise, transport and lower heavy loads. Because of the counterweight it carries to balance these loads, a forklift can weigh twice as much as an automobile."

Common Accidents Due to Misunderstanding:

- Striking objects/pedestrians with swinging rear end
- Tip-overs from excessive speed in turns
- Collisions due to poor visibility
- Injuries from treating forklift like car (jumping out during tip-over)

Slide 46-53: Steering Differences and Training

Rear-Wheel Steering: The Fundamental Difference

How It Works:

- Rear wheels pivot to change direction
- Front wheels (under load) act as pivot point
- Rear end of forklift swings OUTWARD during turn
- Creates "tail sweep" hazard

The Tail Sweep Hazard:

The outward swing of the forklift's rear end is a leading cause of accidents:

- Rear counterweight strikes pedestrians
- Rear hits racking, walls, or equipment
- Operator focuses on front, unaware of rear swing
- Especially dangerous in narrow aisles

Figure-of-Eight Training Exercise:

Forward: "Select gear, do all-round observation check, release handbrake, gently accelerate. Line up pivot point - center of wheel with center of barrel. Glancing back and forth looking at crush point. Gently steer around barrel, maintaining gap two to three inches."

Reverse: "Put transmission in reverse, do observation check all the way around, release handbrake, one hand behind operator seat, look at direction of travel. Get pivot point with centre of wheel with centre of barrel. Feather steering gently, maintaining the gap two to three inches."

Key Steering Skills:

- Smooth control (avoid jerky inputs)
- Observation pattern (check all directions continuously)
- Speed management (slow before turns)
- Pivot point awareness (understand fulcrum)

Turning Safety:

- Reduce speed before entering turn
- Plan turn path (watch front AND rear)
- Turn smoothly and gradually
- Watch overhead and lateral clearances
- Never turn on inclines or ramps while loaded

Section 5: Pre-Operational Checks and Inspections

Slide 54-57: Pre-Operational Checks

Daily Pre-Operational Checklist:

Visual Inspection:

- Check for any visible damage
- Data plate readable and not damaged

- Condition of lift and tilt systems
- Check for cracks in forklift tines
- Each tire for wear, correct inflation, bulge, missing wheel nuts
- Check fuel level

Weekly Checks:

- Hydraulic oil level and hose condition
- Engine oil level
- Transmission oil level
- Brake fluid level
- Cooling water level (when forklift is cold)

Mounting Procedure:

"Before starting any operation, ensure forks are in proper safe position. Recommended width between forks is 1/2 to 3/4s of pallet width. Mount seat from left side holding assist grip with left hand and seat back with right hand. Adjust mirrors and fasten seat belt. Never grab steering wheel or operation levers. Jumping on and off is strictly prohibited."

Starting Procedure:

"Hold steering knob with left hand, confirm select lever is in neutral position, confirm parking brake is set, then insert key and start engine."

Detailed Inspection Process:

"Like any machine, a forklift needs to be in good shape to operate safely. Begin your inspection with a visual walkaround. Look for obvious damage, missing parts, or fuel and oil leaks.

For IC engine forklifts: check fuel, oil levels in crank case and transmission, and coolant.

For electric forklifts: check electrolyte levels in battery (use PPE: eye protection, face shield, rubber gloves). Examine battery terminals for corrosion or loose connections.

Inspect tire condition. Look for damage and remove foreign objects. Verify steering works and moves freely. Check brakes, including parking brake. Examine chain bearings, nuts, and cotters on mast. Lubricate if needed. Ensure hydraulic reservoir has enough oil. Look for leaks. Examine forks for cracks or excessive wear and check alignment.

Ensure headlights, tail lights, turn signals, and flashers function. Test horn, backup alarm, and warning beacon. Verify seatbelts work. If you find anything that isn't right, repair it or report it. Take the forklift out of service if necessary. **Never, ever use a faulty lift.**"

Operational Test:

"Make sure parking brake is on. Listen for unusual noises as engine warms up. Pull lift lever and fully raise and lower forks to ensure free movement. Test side shifter or attachments if installed. Tilt mast back and forth ensuring all hydraulic movements are smooth.

Release parking brake and drive forward or reverse about 20 feet total, ensuring steering and brakes work properly. Check brakes - pedal must not reach floorboard. If it does, brakes are not functioning properly and truck should be taken out of operation.

Turn off engine and apply parking brake. Look under truck to check for oil leaks. Inspect horn, lights and gauges. Complete and file operator checklist and report any defects to supervisor in writing."

Slide 58-59: Data Plate Information

Data Plate Contents:

1. Model number
2. Serial number
3. Mast type
4. Fuel type
5. Back tilt
6. Attachments
7. Front tread
8. Tire size
9. Truck weight
10. Forklift diagram

Electric Forklift Specific:

BATTERY WT. MIN/MAX: Indicates minimum battery weight to meet rated capacity. Maximum weight must not be exceeded or damage could result.

TRUCK WEIGHT W/O BATTERY: Maximum truck weight without battery. Battery weight is stamped on case. Adding truck and battery weight gives total vehicle weight.

Critical Safety Note: Never exceed rated capacity. If attachments are changed or mast height is modified, a new data plate must be requested and installed. Operating without proper capacity information is an OSHA violation.

Refueling and Recharging Safety:

"All refueling and recharging procedures share one critical safety issue: the risk of fire and explosion. Gasoline, diesel fuel, and gases like propane are flammable. Recharging an electric forklift can also be a fire risk because the charging process can cause its battery to generate hydrogen gas, which is also flammable. All refueling and recharging areas need to be well ventilated. There should be no open flames, electrical sparks, or other ignition sources nearby and smoking can never be allowed in these areas."

Slide 60: Forklift Stability

Forklift stability is a function of:

- Center of gravity
- Balance at the fulcrum
- Dynamic forces
- Actual capacity vs rated capacity

Stability Triangle Principle:

"Most forklifts have forks at the front and counterweight at the back, which puts the machine's center of gravity somewhere in the middle. In terms of stability, the vehicle acts like a seesaw. When you put a load on the forks, you add its weight to the forklift and shift the forklift's center of gravity forward.

Draw an imaginary line connecting the two front wheels and the pivot point of the rear axle. This is the stability triangle - it's a map of a forklift's suspension. **To keep the forklift stable, its center of gravity must stay within the stability triangle.** The closer you keep the center of gravity to the middle of the stability triangle, the more stable a forklift will be."

Section 6: Load Handling and Stability

Slide 61-64: Load Handling and Fork Leveling

Step-by-Step Load Handling:

"Approach your load in a straight line, targeting the center of the pallet for accurate alignment. Stop the forklift a few centimeters before the load, and set up your forks parallel—never pick up a load with tilted or uneven forks. Make sure the middle section of the fork lines up with the middle of the pallet.

Enter slowly, observing fork position continuously. If the forks touch the pallet or are misaligned, stop and reset. Once positioned, lift the load gradually, ensuring it is balanced and secure. Begin backing out slowly, continuing to check the load and surroundings. Lower the load to the lowest safe travel height (4-6 inches) above the floor and tilt the mast slightly backward before traveling."

Fork Leveling Technique:

"Learning to level the forks is one of the most important skills for picking up and placing loads. For new operators, judging fork level from the cab can be difficult—never hesitate to step off to take a closer look.

Testing fork level: Observe pallet movement when it's first lifted:

- If the far end moves before the close end, forks are tilted too far back
- If the front moves before the rear, forks are tilted too far forward
- **When both ends move together, forks are perfectly level**

When entering the load, approach straight. When picking up, nudge the load gently so the forks are engaged. Lift only two to six inches, just enough for clearance. In real-world operation, always tilt the mast back after lifting. This secures the load's center of gravity and greatly increases safety."

Additional Load Handling:

"To pick up a load, position the forklift squarely with the pallet. Drive forward slowly, sliding the forks into the pallet until it touches the vertical portion of the forks or the back rest. The length of the forks should be 2/3 the length of the load at a minimum. Center the load side to side whenever possible. Raise the load only as much as you need to. Use only enough backward tilt to stabilize it. Before you move off, lower the forks to between 4 and 6 inches off the ground. **If the load obstructs your forward vision, operate the forklift in reverse with the load trailing** so you have a clear view in the direction you're traveling."

Stability Triangle and Tip-Over Prevention:

"If the load on the forks weighs less than the machine, the forklift center of gravity will be behind the front wheels and it will still be stable. If the load and truck weigh the same, the forklift's center of gravity will be on the front wheels—this means the forklift could easily become unstable, for instance if you had to stop quickly. If the load weighs more than the forklift, the center of gravity ends up in front of the wheels, the back wheels will come off the ground and the forklift will tip forward."

Slide 65-71: Load Capacity and Load Charts

Load Center and Capacity:

"Every forklift has a maximum weight it can lift, known as its load capacity. The load capacity will also indicate how far back on the forks a load's center of gravity should be for the truck to lift the load safely. This distance from the vertical part of the forks or the back rest to the load's center of gravity is known as the load center.

The load center for most forklifts is 24 inches (half the depth of a standard pallet), but loads can come in different shapes and sizes with different load centers. So, while a forklift may be rated to safely lift 4,000 pounds at a 24-inch load center,

if you pick up a 4,000-pound load that has a 36-inch load center, it will shift the forklift's center of gravity further to the front and cause it to tip forward. **For every 1 inch you lose about 114 lbs of actual capacity!**"

Load Charts (Telehandlers/Rough Terrain):

"A load chart is critical for safe telehandler and forklift operation. It tells the operator how much the machine can lift at any given point. Load charts are always model and attachment specific, so you need to use the right chart every time.

Load charts indicate the 'load center'—half the distance of the forks—because this changes how much weight you can safely lift. The chart will also show the boom extension, angle indicators (arc, degrees), reach distance (out), lift height (up), and combined bands indicating lifting limits.

Physical boom extension and angle indicators on the telehandler help you reference the chart from the operator's seat. For example, if you need to place a load 35' high but are 25' back from the scaffolding, find 25' out and 35' up on the chart—if the chart says 2,000 pounds, that is the max safe load for that position."

Load Weight Calculation:

"Figuring out the weight of the load you're lifting is one of the first things you should do. The total weight must account for every piece of lifting gear involved: hook block, lifting beams, shackles, slings, and other rigging hardware.

Methods to determine load weight:

1. Check if weight is marked on the load
2. Reference engineered prints, product specifications, manufacturer's documentation
3. Use load measuring devices or scales when available
4. Calculate the weight yourself

Calculation methods:

- For rectangular/square objects: Volume = Length × Width × Height
- For hollow cylinders: Volume = 3.14 × Length × Wall Thickness × (Diameter – Wall Thickness)
- For complex shapes: Break into smaller rectangles, calculate each part, then add together

Next, use a material weight table to find pounds per cubic foot. Multiply material density by calculated volume to derive total weight.

Examples:

- Aluminum block (6' × 3' × 4'): Volume = 72 ft^3; Weight = 72 × 165 = 11,880 lbs (5.94 tons)
- Steel pipe (8' L, 3' OD, 1.5" wall): Convert 1.5" to feet (0.125'), calculate: 3.14 × 8 × 0.125 × 2.875 = 9.03 ft^3; Weight = 9.03 × 480 = 4,334 lbs (2.17 tons)"

Slide 72-74: Rules of the Road and Common Errors

Safe Operating Procedures:

"Operating a forklift safely begins before you even climb onto the machine. It starts with you asking yourself if you're ready to drive. If your back is giving you trouble, you've had an argument with a coworker, or you're taking medication for a head cold, be careful. **Never drive a forklift if you're under the influence.**

Buckle up before turning the key. Look around, ensure the way is clear, follow normal rules of the road: keep to the right, give pedestrians right of way, always face the direction you're traveling, keep arms/legs inside the cab, keep the forklift under control at all times. Jerky and erratic driving can cause accidents. Never horseplay.

With or without a load, drive with forks lowered to 4-6 inches off the ground and mast tilted back. Don't drive faster than a walking speed."

Common Operator Errors:

"The first critical step is to ensure you are wearing a seatbelt that is in good condition and fully functional before moving the forklift. Always check behind, in front, and to the sides for pedestrians or obstacles before moving in any direction—situational awareness must be a constant priority.

Critical mistakes to avoid:

- Not wearing seatbelt
- Not checking behind before reversing
- Keeping limbs outside operator compartment
- Driving at unsafe speed or faster than facility conditions allow
- Traveling with load that blocks view (travel in reverse instead)
- Carrying unsecure loads
- Using cell phones or being distracted while operating
- Executing abrupt operations (acceleration, braking, turning)
- Operating mast while forklift is moving

Do not execute abrupt operations: Operation of the mast while the forklift is moving is strictly prohibited. It is important to be calm and follow each operational step even during busy conditions."

Stability and Balance Video Key Points:

"Over one-quarter of all lift truck fatalities occur when a lift truck tips over. Understanding the principle of lateral stability will reduce the chance of your truck falling sideways.

Is a lift truck more stable with a load or without a load while turning?

As a lift truck turns with a load, the combined center of gravity moves towards the edge of the triangle. When turning without a load, notice the center of gravity is much closer to the edge of the triangle. The closer it is, the less it takes for the truck to become unbalanced and tip over. **This is why an unloaded truck should be operated just as carefully while cornering.**

Causes of tip-overs: sharp turns, raised masts, unstable loads, potholes, uneven surfaces, wet surfaces, and driving on ramps."

Section 7: Safe Operation and Pedestrian Safety

Slide 75-79: Pedestrian Safety and Riders

OSHA Requirements (29 CFR 1910.176(a) and 1910.178):

- Pedestrian walkways must be clearly marked
- Adequate lighting required in all operating areas
- Physical barriers between forklift traffic and pedestrian areas when feasible
- Audible and visual warnings on all forklifts (horns, lights, alarms)

Pedestrian Right-of-Way Rules:

- **Pedestrians ALWAYS have right-of-way** unless in designated forklift-only zones
- Operators must yield to pedestrians at all intersections
- Operators must establish eye contact before proceeding
- Operators must sound horn when approaching blind corners

Statistics:

"Pedestrians were involved in about 40% of all lift truck accidents - 18% of them are fatal. When traveling, it may be difficult for you to see through the mast and/or around the load. Pedestrians may not hear you due to noise or because they are concentrating on their job."

No Riders Policy:

"Lift trucks are strictly designed for one person - the operator. **No riders ever.** Never lift a person using a lift truck - not on the forks, not in a cage, and not even in a so-called OSHA-approved man lift basket. 12% of fatalities related to lift trucks involve someone using a lift truck as an elevator.

Don't let anyone walk under raised forks, whether they're loaded or not. Don't let co-workers ride on your forklift either. And never let anyone ride on the load."

Personnel Platform Safety:

"A forklift can be very convenient for raising co-workers up high when they need to perform maintenance or other tasks off the ground, but only if you do it safely. Never lift someone who's standing on the bare forks or on a pallet. To lift people safely, you need to use an aerial platform, also known as a cage. Make sure the platform is firmly against the vertical part of the forks or the back rest. Then secure it with a safety chain. **Never move a forklift with the platform elevated or with someone on it.** And always stay with the truck when the platform is raised."

Sources: OSHA pedestrian safety standards[35][38][41][47][50][53]

Slide 80-82: Tip-Over Procedures

What to Do If Forklift Tips Over:

"No matter how careful you are with your forklift, accidents can still happen. You need to know what to do if your forklift tips over. **Do not try to jump out. Studies show that will probably get you killed.**

Proper tip-over response:

1. **Brace your feet** against the floor
2. **Grab onto the steering wheel** and pull yourself tight against it
3. **Lean in the opposite direction** from the way the vehicle is tipping
4. **Hang on** and stay in the cab

Why this works: The overhead guard protects you. If you jump, the overhead guard will strike and crush you. Staying inside the protective cage is your best chance for survival."

Tip-Over Story:

"On October 17th, 1990, Scott was operating a lift truck outside. He was attempting to pick up a load of empty pallets while making a sharp turn. He hit a small pothole.

Scott was fortunate - he had been properly trained. He knew how to best survive a tip-over situation. First of all, Scott was wearing a seatbelt. When the lift truck tipped, he stayed inside the operator compartment - his best chance for survival. He braced his feet and held on tight to the steering wheel with both hands while leaning in the opposite direction of the fall. Scott was able to escape with minor injuries.

Never attempt to jump out of your lift truck if you encounter a tip-over. Had Scott tried to jump clear of the truck, the overhead guard would have struck him, crushing him to death."

Section 8: Loading Docks and Parking

Slide 83-84: Loading Dock Safety

Loading Dock Hazards:

"Loading docks are busy places for pedestrians and equipment. On average, a lift truck may pass over a dock 100,000 times per year, so there are plenty of chances for accidents. Statistics show that over 7% of accidents happen in the loading dock area.

If the area between the trailer and the dock is not safely secured for any reason, the lift truck can fall through the gap, causing injury or death."

Common Hazards:

1. Early Departure: Tractor-trailer driver assumes loading/unloading is complete and pulls away while lift truck is entering or leaving trailer. Even wheel chocks offer little protection if driver leaves early.

2. Landing Gear Collapse: When tractor is removed, if landing gear is weak or corroded, it may collapse due to weight and motion of forklift and trailer.

3. Trailer Tipping: Heavy loads placed in nose of trailer exceed counterbalance effect of trailer's rear axle. Uncoupled trailers must be secured with support jack before entering.

4. Trailer Creep: As lift truck travels on and off trailer, its momentum inches trailer away from dock. This is the most common form of trailer separation. Tractor-trailer brakes must be engaged and wheel chocks properly installed. However, wheel chocks have limitations and could fail if placed on ice, gravel or sand.

Safety Solutions:

Wheel Chocks: Triangular, wedge-shaped objects made of hard rubber inserted in front of trailer's rear wheels. "But you can't rely solely on wheel chocks to prevent a trailer from moving because they sometimes fail. On slippery pavement, wheel chocks can slide when the truck attempts to move. Or the lift truck moving in and out of the trailer can cause trailer creep."

Mechanical Restraint Systems: Permanently installed device at outside of loading dock with hook that connects to trailer's ICC bar (lower metal bar at back end of trailer). This device will keep trailer locked in place if driver tries to leave prematurely.

Signal Lights: Situated both inside and outside of loading dock. When light is red outside, it flashes green inside—signal to forklift operator that it is safe to enter trailer. When outside light flashes green, driver knows it is safe to check in before departing.

Glad Hand Lock: Disconnecting the air hose between tractor and trailer, preventing hoses from being connected. This releases air from trailer brakes, essentially locking them.

Key Exchange: Driver gives keys to warehouse worker. When loading/unloading complete, keys are returned.

Other Essential Points:

- Inspect inside of trailer to ensure floor is not damaged and capacity will support weight of load, lift truck, and operator
- Secure trailers detached from tractor with support jack at front end before lift truck enters (prevents trailer from tipping upwards when load is removed from front)

Dock Plate Safety:

"Whether you're loading or unloading, always use a dock plate to bridge the gap between trucks and rail cars and the loading dock. Make sure the plate is in good condition and secured firmly in place. Verify that it's rated to handle the

weight of your forklift plus the weight of the load you will be carrying. And drive across the dock plate slowly and carefully."

Slide 85: Forklift Maintenance Requirements

Daily Pre-Operation Inspection (Required by OSHA):

Every operator must conduct and document daily inspection before each shift. If defects are found, forklift must be removed from service until repaired.

Service Intervals for Internal Combustion Forklifts:

Every 250-300 hours (approximately every 6 weeks):

- Change oil and filters
- Check and adjust engine idle speed and timing
- Lubricate chassis and mast components
- Blow out and inspect brakes
- Inspect hydraulic system and mast
- Check belts, hoses, drivetrain, steering
- Inspect tires for damage and wear
- Inspect electrical system
- Check coolant, transmission fluid, brake fluid

Every 500-600 hours (approximately every 4 months):

- All 250-hour service items
- Inspect chains for damage and tension
- Check mast operation and carriage rollers
- Inspect lift and tilt cylinders
- Adjust chassis links and clutch
- Inspect hydraulic pump and change hydraulic filter
- Clean radiator
- Check pedal free play and adjust hand brake

Annually or 2,000 hours:

- Major inspection of all systems
- Replace major wear components as needed
- Comprehensive safety inspection

Electric Forklifts:

Every 500 hours (approximately every 12 weeks):

- Check motor and controller
- Inspect cables and connections
- Lubricate moving parts
- Inspect and test battery (electrolyte levels, terminals, charging system)
- Check lights, accessories, and attachments
- Conduct safety inspection

Documentation Requirements:

- Maintain service logs for each piece of equipment
- Track hours of operation
- Record all repairs and parts replaced
- Document inspection findings and corrective actions
- Retain records per OSHA and manufacturer requirements

Sources: OSHA maintenance requirements[36][39][42][45][48][51]

Slide 86-87: Parking a Forklift

Proper Parking Procedure:

"When your shift is over, or anytime you're going to leave a forklift unattended, be sure to secure it. Lower the forks to the ground, set the parking brake, turn off the motor, and take the keys with you. If you have to park on an incline, chock the wheels to make sure the forklift stays where you left it.

Step-by-step parking:

1. Depress the brake pedal
2. Stop the forklift on level ground
3. Apply the parking brake
4. Shift to neutral
5. Tilt the mast forward until the forks touch the ground
6. Turn the key switch off
7. Remove the key
8. Get off using proper mount/dismount technique
9. Store the key securely

Never:

- Block doors or emergency exits
- Park on inclines without wheel chocks
- Leave forklift unattended with engine running
- Leave forks elevated
- Park in pedestrian walkways"

Section 9: Attachments and Equipment-Specific Training

Slide 38: Ramps, Slopes, and Inclines

OSHA Requirements (29 CFR 1910.178(n)(7)):

- All grades must be ascended and descended slowly
- For grades over 10%, loaded forklifts must travel with load upgrade (facing uphill)
- Tilt load back and raise only enough to clear ground
- Maintain safe distance from ramp edges
- Do not turn on ramps or inclines
- Always look in direction of travel

Calculating Ramp Grade:

Grade % = (Rise ÷ Run) × 100
Example: 5 ft rise ÷ 25 ft run = 20% grade

Loaded Forklift Procedures:

- **Ascending:** Drive forward with load uphill, forks elevated 4-6 inches, mast tilted back
- **Descending:** Drive in reverse with load uphill (trailing behind), maintain clear view by turning head

Unloaded Forklift Procedures:

- **Ascending:** Drive backward up the ramp with forks pointed downgrade
- **Descending:** Drive forward down the ramp with forks pointed downgrade

Additional Safety Measures:

- Check ramp condition before use (no damage, debris, or slippery surfaces)
- Verify ramp capacity supports combined weight of forklift and load
- Use spotter if visibility is limited
- Never exceed manufacturer's maximum grade specification
- Keep away from edges—running one wheel off can cause tip-over
- Complete all turns before entering or after exiting ramp

Maximum Recommended Grades:

- OSHA construction standard limits ramps to 20 degrees (36% grade)
- Most forklift manufacturers specify maximum gradeability in operator manual
- Grades exceeding 10% require special precautions

Sources: OSHA ramp and grade standards[34][37][40][43][46][49][52]

Slide 88: Forklift Attachments

Common Types of Attachments:

- **Side shifters:** Allows left/right movement of load for precise placement
- **Fork positioners:** Adjusts fork width from cab for different-sized pallets
- **Fork rotators:** Rotates forks for dumping or handling bins
- **Hinged fork attachments:** Enables movement of forks for specialized handling
- **Push-pull attachments:** Used for loads without pallets
- **Multi-pallet handlers:** Carries two or more pallets at once for greater efficiency
- **Boom attachments:** Telescoping/non-telescoping arms for irregular objects or construction materials
- **Clamps:** Paper roll, bale, carton, drum, pipe clamps secure specialty items

Functions & Use Cases:

- For precise stacking, side shifters and positioners are essential
- Rotators make dumping tasks possible
- Boom attachments and clamps allow handling of items outside standard pallets
- Multi-pallet handlers improve productivity in warehouses

Safety & Capacity Notes:

- **Always confirm forklift's rated capacity with attachment installed—capacity typically decreases**

- Every attachment requires thorough daily pre-use inspection for leaks, cracks, or improper connection
- Only operators specifically trained and evaluated for each attachment type may use them

OSHA Requirements:

- OSHA mandates hands-on, attachment-specific training and documentation before an operator can use a new attachment
- Refresher training must be provided if a new equipment type or attachment is introduced to the workplace

Best Practice Example:

When using a multi-pallet handler, adjust forks to correct width, travel slowly in tight aisles, and watch for tip-over risk when carrying wide or uneven loads.

Slide 89: Equipment-Specific Training

OSHA Mandate:

Training must be specific to:

1. The type of powered industrial truck the operator will use
2. The actual workplace conditions and hazards present

Equipment-Specific Training Includes:

- Controls and their functions for specific make/model
- Capacity ratings and load charts for that equipment
- Visibility limitations unique to that forklift
- Attachment-specific operation and capacity changes
- Manufacturer's operating procedures
- Maintenance and inspection requirements
- Emergency procedures specific to equipment type

When New Training Required:

- Operator assigned to different class/type of forklift
- Different fuel type (electric to propane)
- Addition or change of attachments
- Significantly different make/model with different controls
- Moving to facility with different hazards or layout

Workplace-Specific Training Includes:

- Facility layout, aisle widths, and traffic patterns
- Loading dock procedures and dock plate use
- Ramp and incline locations and grades
- Surface conditions (smooth vs. rough terrain)
- Pedestrian traffic areas and crossing points
- Overhead clearances and obstructions
- Temperature and ventilation considerations
- Refueling/recharging locations and procedures
- Emergency exits and procedures

- Facility-specific speed limits and traffic rules

Note: An operator certified at one facility is NOT automatically qualified at another facility—site-specific training and evaluation are required[14][15][27].

Summary and Key Takeaways

Review of Critical Safety Points

Training and Qualification:

- Operators must be 18+, trained, evaluated, and certified
- Refresher training required every 3 years minimum
- Site and equipment-specific training mandatory

Daily Pre-Operation:

- Inspect forklift before every use
- Report and repair defects immediately
- Never operate faulty equipment

Safe Operation:

- Always wear seatbelt
- Travel with forks 4-6" off ground, mast tilted back
- Drive at safe speeds (walking pace in congested areas)
- Check all directions before moving
- Keep arms and legs inside operator compartment
- No cell phones or distractions while operating

Load Handling:

- Know and follow capacity ratings
- Ensure loads are stable and secure
- Level forks before entry
- Lift only high enough for clearance
- Travel in reverse if load blocks vision

Stability and Tip-Overs:

- Keep center of gravity within stability triangle
- Never exceed rated capacity
- Do not turn with elevated loads
- Watch for surface conditions
- If tip-over occurs: Stay in cab, brace, lean away from fall, hold on

Pedestrian Safety:

- Pedestrians always have right-of-way
- Sound horn at intersections
- Make eye contact before proceeding
- No riders ever (except in approved personnel platforms)

Ramps and Inclines:

- Loaded forklifts: load facing uphill at all times
- Unloaded forklifts: forks facing downgrade
- Travel slowly, no turning on ramps

Parking and Securing:

- Lower forks, neutral, parking brake, engine off, remove key
- Chock wheels if on incline
- Never block exits or fire lanes

Questions?

When in doubt, ask your supervisor. It's better to take extra time to do it safely than risk injury, damage, or worse.

References

[2] OSHA Forklift Checklist 2025
[3] OSHA Powered Industrial Trucks Overview
[12] OSHA eTool Training Assistance
[14] OSHA Forklift Certification Requirements & Training Rules
[15] OSHA's Forklift Training FAQs - JJ Keller
[16] Forklift Certification Training - OSHA Education Center
[17] OSHA Standard 1910.178 - Powered Industrial Trucks
[20] OSHA Powered Industrial Trucks Hazards and Solutions
[21] Formal Hazard Assessment - Operating a Forklift
[22] How to Become a Forklift Operator - Aerotek
[23] Top 12 Forklift Safety Hazards & How To Avoid Them
[24] OSHA Forklift Operator Training Requirements - HSI
[27] OSHA Interpretation - Training Content and Certification
[28] Forklift Fundamentals - U.S. Army Combat Readiness Center
[29] OSHA Safety Management - Hazard Identification
[30] Key OSHA Forklift Training Rules - Toyota MHS
[31] 6 Important Forklift Safety Hazards - BigRentz
[32] OSHA Forklift Trainer Requirements - Conger
[33] Forklift Safety: Hazards & Prevention - OSHA.com
[34] Forklift Ramp Slope Requirements - BigRentz
[35] Forklift Pedestrian Safety – OSHA Requirements
[36] Forklift Maintenance Guide - Conger
[37] OSHA eTool: Ramps and Grades
[38] Forklifts in the Workplace – Protecting Pedestrians
[39] OSHA Vehicle Maintenance Standards
[40] Safe Forklift Use on Ramps - Toyota Forklifts
[41] How to Stay Safe Around Forklifts - OSHA Education Center
[42] Forklift Maintenance Schedule - Toyota MHS
[43] OSHA Guidelines for Driving on Ramps - BHS
[45] Forklift Maintenance - Maintenance Care
[46] Essential Guidelines for Dock Ramp Slopes
[47] Improve Pedestrian and Forklift Safety - Voxel
[48] Forklift Maintenance Best Practices - Raymond
[49] Operating Forklifts On Inclines - Logisnext
[50] Forklift and Pedestrian Safety - Arbill
[51] Forklift Preventive Maintenance Checklist - Limble CMMS

[52] Safety Precautions on Ramps - Forklift Safety
[53] Forklift Pedestrian Safety - KPA
[54] What is a Counterbalance Forklift - Inbound Logistics
[55] Narrow Aisle Forklifts - Toyota Forklift
[56] Forklift Overhead Guard Assemblies - Morrison Industries
[57] What is a Counterbalance Forklift - Adaptalift
[58] Very Narrow Aisle Forklifts - Raymond
[59] Parts of a Forklift and Their Functions - Conger
[61] Narrow Aisle Forklift Types - BigRentz
[62] Anatomy of a Forklift - Logisnext
[65] Purpose of Overhead Guard - Eureka Forklifts
[68] 10 Most Important Forklift Components - Ring Power
[70] Narrow Aisle Forklifts - Forklift Certification
[71] Forklift Anatomy - G&W Equipment
[72] What is a Counterbalance Forklift - Masterlift Ireland
[73] Very Narrow Aisle Trucks - Yale
[74] Guide to Forklift Operator Training
[75] Powered Industrial Truck Safety Instructor Workbook
[77] Forklift Safety Manual - Safe At Work California
[78] Forklift Training Manual - Safety Culture
[80] Forklift Safety Guide - Washington State L&I
[81] Forklift Operator's Handbook - RTA Dubai

Document End

This comprehensive training manual provides complete OSHA-compliant forklift training content covering all essential topics for powered industrial truck operation. Use this material for instructor-led training, student reference, and compliance documentation.

For questions or additional training resources, contact:

Mandy Richardson
ANEW - Building People, Building Communities
Email: Mandy@anewcareer.org
Phone: 206-710-1019